

Images and evidence should be included throughout.

The manager should make a copy, share it with each member of the group (jmiranda@rpsk12.org, pnguyen@rpsk12.org), make it viewable by “ANYONE with the link” and submit it to Google Classroom when complete. Delete the notes in red when you are done with each section.

Group Members: Nizar, Vitoria, Rocio, Mariana, Denia, Mihailo

Element A: Presentation and Justification of the Problem

Item	Notes/Links/Details
Problem Statement: <i>Improving the level of understanding of parabolas in Freshmen math through an interactive hands on experience that includes critical thinking.</i>	
Problem Definition: Freshmen don't understand parabolas because they do not remember the concepts taught to them, lowering their engagement in the lessons. This often occurs due to the lack of understanding of the words. Math terms can be difficult concepts if not explained thoroughly, repeated multiple times, and emphasized. Therefore, for the completion of problems containing parabolas are not successful.	
Additional information about the problem: This problem impacts other stakeholders because if they do not understand the concept of parabolas taught during freshman year their mathematical abilities in the following years may be harmed.	

Stakeholders: Freshmen who struggle with parabolas. Freshmen who do not ask for help. Upperclassman (Juniors/Seniors) Teachers MCAS data	
Concerns of Stakeholders: Freshmen could end up failing quarters in math class which can lead to ongoing failure in the upcoming years. Freshmen math is the base of all math classes in high school and failing it could backtrack the students and make them stay behind while other students continue their math learning. (they may need to retake the class during sophomore year or junior year) A concern for teachers is having a disengaged class where no students are participating. Other grade level teachers would have to fill in the students who don't understand again. Concerns of upcoming juniors and seniors who have struggled with freshman math head on a path of forgetting the meaning of mathematical terms which then makes it harder on the student and stresses over the definition and work that should be done.	
Problem Justification: Previous studies from the University of South Carolina have shown that many incoming freshman students are not as prepared for word problems when coming into Algebra One. Although they may be able to solve computational problems they struggle with word problems. "On top of this students will usually avoid the question or use simple formulas without understanding the concept of the question".	

Element A: Presentation and Justification of the Problem

	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A1. Problem identification is _____.	very clear and objective	clear and objective	somewhat clear and objective	somewhat clear or objective	unclear	missing (cannot be inferred from information included)
A2. Problem definition is _____.	in considerable depth	in more than adequate depth	in adequate depth	in superficial depth	unclear	missing (cannot be inferred from information included)
A3. Problem elaboration is _____.	well elaborated	elaborated	imprecisely or generally elaborated	minimally elaborated	unelaborated and/or is subjective	not elaborated (cannot be inferred from information included)
A4. The concern of primary stakeholder groups. Justification of the problem highlights the concerns of _____.	almost all primary stakeholder groups	many primary stakeholder groups	several primary stakeholder groups	at least a few primary stakeholders/stakeholdersolder groups	only one or two stakeholders/stakeholdersolder groups	no primary stakeholders/stakeholdersolder groups
A5. Problem justification is based on _____.	numerous, timely, credible, and various sources	timely, credible, and various sources	credible and timely sources	credible sources	partially-credible sources	essentially only the opinion of the researcher
A6. Problem justification offers _____ -objective detail.	consistently	mostly	generally	somewhat	unclear	no